

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

A Research Paper Submitted as Part of the Unified Mentor Internship Program

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Live Dashboard: <https://shipping-route-efficiency-analysis-cww.streamlit.app/>

Repository: <https://github.com/Saswata-pal/shipping-route-efficiency-analysis>

Abstract

Nassau Candy Distributor operates a nationwide distribution network that moves confectionery products from five regional factories to customers across the United States. Although the organization captures granular order- and shipment-level data, this information is not currently transformed into route-level operational intelligence. As a result, logistics decisions are made reactively rather than proactively. This paper presents a structured, data-driven methodology for evaluating factory-to-customer shipping route efficiency. The study defines routes as factory-to-customer-state/region pairs, engineers shipping lead time as the core performance metric, and benchmarks routes, regions, and shipping modes against one another to surface bottlenecks and best practices. The paper further documents the design of an interactive Streamlit dashboard that operationalizes these findings for ongoing monitoring, and closes with recommendations intended to move Nassau Candy from reactive to data-driven logistics decision-making.

Keywords: logistics analytics, route efficiency, shipping lead time, supply chain optimization, geographic bottleneck analysis, Streamlit dashboarding

1. Introduction

1.1 Background

Nassau Candy Distributor is a national distributor of confectionery products, shipping goods manufactured at five factories to customers spread across multiple U.S. regions. In distribution

operations of this scale, shipping efficiency is not a peripheral concern — it directly shapes customer satisfaction, operating cost, and the organization's ability to scale into new markets. Delayed shipments increase handling and expediting costs, strain customer relationships, and mask the underlying causes of network inefficiency. Despite this, many distributors — Nassau Candy included — continue to make routing and carrier decisions based on institutional habit rather than systematic analysis of historical shipment performance.

1.2 Problem Statement

The organization currently lacks visibility into several fundamental logistics questions:

- Which factory-to-customer routes are consistently efficient, and which are chronically slow?
- How does shipping performance vary by region, state, and shipping mode?
- Where do operational bottlenecks concentrate geographically?
- Which combinations of high shipment volume and poor performance represent the greatest opportunity for improvement?

Without answers to these questions, logistics optimization at Nassau Candy remains reactive — problems are addressed only after customers complain or costs spike, rather than being anticipated and managed proactively.

1.3 Objectives

This study aims to:

1. Establish a repeatable methodology for measuring shipping route efficiency using existing order and shipment data.
2. Identify the best- and worst-performing factory-to-customer routes.
3. Quantify how shipping mode choice affects delivery speed and reliability.
4. Surface geographic bottlenecks where volume and delay coincide.
5. Translate these findings into an interactive dashboard that supports ongoing, self-service monitoring by logistics stakeholders.

2. Data Description

2.1 Order and Shipment Dataset

The analysis draws on order-level shipment records containing the following fields:

Field	Description
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Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date the order was placed
Ship Date	Date the order was shipped
Ship Mode	Shipping method used for the order
Customer ID	Unique customer identifier
Country/Region	Country or region of the customer
City	Customer city
State/Province	Customer state or province
Postal Code	Customer postal/zip code
Division	Product division
Region	Customer region
Product ID	Unique product identifier
Product Name	Full product name
Sales	Total sales value of the order
Units	Total units ordered
Gross Profit	Sales minus cost
Cost	Manufacturing cost

2.2 Factory Network

Nassau Candy's production network consists of five factories, each geographically anchored and each specializing in a subset of product divisions:

Factory	Latitude	Longitude
Lot's O' Nuts	32.88189	-111.768036

Wicked Choccy's	32.07617 6	-81.088371
Sugar Shack	48.11914	-96.18115
Secret Factory	41.44633 3	-90.565487
The Other Factory	35.1175	-89.971107

The factories are geographically dispersed across the Southwest (Lot's O' Nuts, Arizona), Southeast (Wicked Choccy's, Georgia), Upper Midwest (Sugar Shack, North Dakota/Minnesota border), Midwest (Secret Factory, Illinois/Iowa border), and Mid-South (The Other Factory, Tennessee/Mississippi border) regions. This dispersion is significant for route analysis: a customer's distance-driven lead time will vary substantially depending on which factory produces the product they ordered, independent of shipping mode.

2.3 Product-to-Factory Mapping

Each product division is manufactured at a specific subset of factories:

Division	Product Line (examples)	Factory
Chocolate	Wonka Bar – Nutty Crunch Surprise, Fudge Mallows, Scrumdiddlyumptious	Lot's O' Nuts
Chocolate	Wonka Bar – Milk Chocolate, Triple Dazzle Caramel	Wicked Choccy's
Sugar	Laffy Taffy, SweeTARTS, Nerds, Fun Dip	Sugar Shack
Other	Fizzy Lifting Drinks	Sugar Shack
Sugar	Everlasting Gobstopper	Secret Factory
Other	Lickable Wallpaper, Wonka Gum	Secret Factory
Sugar	Hair Toffee	The Other Factory
Other	Kazookles	The Other Factory

Because product and factory are tightly coupled, every order can be traced back to a single point of origin, allowing the construction of true factory → customer-state/region routes rather than relying on distributor-level averages.

3. Methodology

3.1 Data Cleaning and Validation

Before any route-level analysis is performed, the raw order/shipment data is cleaned to ensure integrity:

- **Date validation:** Order Date and Ship Date fields are parsed and validated for correct formatting; records with unparseable dates are flagged.
- **Lead time sanity checks:** Any record where Ship Date precedes Order Date (implying a negative lead time) is treated as a data-entry error and removed or corrected.
- **Missing shipment handling:** Orders with missing Ship Date or Ship Mode values are excluded from lead-time calculations but retained for sales/volume analysis where appropriate.
- **Geographic standardization:** State/Province and Region fields are standardized (e.g., consistent abbreviations, correction of misspelled city/state names) to prevent the same route from being fragmented into multiple inconsistent labels.

3.2 Feature Engineering

Two engineered features underpin the entire analysis:

1. **Shipping Lead Time** = Ship Date – Order Date, measured in days. This is the primary unit of efficiency measurement throughout the study.
2. **Route Identifier**, constructed at two levels of granularity:
 - *Factory* → *Customer Region* (coarse-grained, useful for high-level trend spotting)
 - *Factory* → *Customer State* (fine-grained, useful for pinpointing specific bottlenecks)

Shipments are additionally grouped by Ship Mode (e.g., Standard, Expedited) to allow performance to be evaluated independently of, and in combination with, geography.

3.3 Route Definition and Aggregation

Each unique factory-to-customer-state/region pair is treated as a distinct route. For every route, the following are aggregated:

- **Total shipment volume** — the number of orders fulfilled on that route.
- **Average shipping lead time** — the mean number of days from order to ship.
- **Lead time variability** — the standard deviation of lead time, which captures consistency (or lack thereof) independent of the average.

A route with a low average lead time but high variability is operationally different — and arguably more problematic for planning purposes — than a route with a moderate but highly consistent lead time.

3.4 Efficiency Benchmarking

Once routes are aggregated, they are ranked from fastest to slowest average lead time. This ranking supports:

- Identification of the **Top 10 most efficient routes** — candidates for best-practice case studies.
- Identification of the **Bottom 10 least efficient routes** — priority targets for intervention.
- Cross-comparison of performance across ship modes, to determine whether slow routes are a function of geography, carrier choice, or both.

3.5 Geographic Bottleneck Analysis

Beyond route-level ranking, the analysis identifies broader geographic patterns by flagging:

- Regions or states with high average lead time regardless of route.
- Regions where high shipment volume coincides with poor performance — these represent the highest-impact bottlenecks, since they affect the largest number of customers.
- Congestion-prone states that appear repeatedly among the slowest routes across multiple factories.

3.6 Ship Mode Performance Analysis

Shipping mode is analyzed as an independent variable to evaluate cost-time tradeoffs. Standard and Expedited shipping are compared on:

- Average lead time differential.
- Consistency (variability) of delivery time.
- Whether the incremental speed of Expedited shipping is proportionate to its (implied) cost premium, evaluated descriptively rather than through a formal cost model, since per-shipment carrier cost is not directly captured in the dataset.

3.7 Technical Implementation and Tools

The methodology above was implemented as a full analytics pipeline rather than a one-off analysis, using the following stack and techniques:

- **Python** as the core language, with **Pandas** and **NumPy** used for data cleaning, date parsing, lead-time computation, and route-level aggregation (**groupby/agg** pipelines across factory–state and factory–region keys).

- **A staged notebook pipeline**, splitting the workflow into nine discrete, auditable steps — data loading, cleaning/validation, feature engineering, route definition/aggregation, efficiency benchmarking, geographic bottleneck analysis, ship-mode performance analysis, key insights/recommendations, and final export — so each transformation could be validated independently before being passed downstream.
- **Route Efficiency Score normalization**: rather than ranking routes on raw average lead time alone (which penalizes long-haul routes unfairly), lead times were rescaled to a 0–100 normalized score, where a higher score indicates faster/better relative performance. This makes short regional routes and long cross-country routes comparable on the same scale.
- **Rule-based bottleneck detection**: geographic bottlenecks were flagged programmatically as states/regions falling simultaneously in the top 25% of shipment volume and the top 25% of average lead time — surfacing the intersection of "high impact" and "high delay" rather than relying on either metric alone.
- **Plotly (Express and Graph Objects)** for all visualizations, including bar charts, scatter plots (volume vs. lead time), and a **US choropleth map** for state-level lead-time visualization.
- **Streamlit** for the interactive dashboard layer, with a custom CSS theme applied on top of the default Streamlit styling, and a responsive filter panel (sidebar on desktop, collapsible expander on mobile) so the same dashboard works across device sizes.
- **Dynamically generated insights**: recommendation text shown on the dashboard (e.g., in the Route Drill-Down and Trends views) is computed on the fly from whatever subset of data is currently filtered, rather than hardcoded against the full dataset — so the narrative insights stay accurate as a user changes date range, region, ship mode, or delay threshold.
- **Dual export paths**: cleaned and feature-engineered data was exported both to a Streamlit-ready processed format and to a separate Power BI-ready format, so the same underlying analysis could be consumed outside the Python/Streamlit environment if needed.
- **Cloud deployment** via Streamlit Community Cloud, connected directly to the project's GitHub repository so that new commits to the deployment branch automatically redeploy the live dashboard.

4. Key Performance Indicators

The study is organized around five core KPIs:

KPI	Definition
Shipping Lead Time (Median)	Median Ship Date – Order Date across shipments, used alongside the mean to reduce sensitivity to outlier delays

Average Lead Time (Mean)	Mean shipping duration per route
Route Volume	Average / total number of shipments per route
Delay Frequency	Percentage of shipments exceeding a configurable lead-time threshold
Route Efficiency Score	Normalized (0–100) lead-time performance, scaled for cross-route comparison — higher is faster/better
Total Shipments / Unique Orders	Overall network throughput, and average shipments per order

The Route Efficiency Score in particular allows routes of very different volume and geography to be compared on a common scale, rather than relying on raw lead time alone, which can be misleading when comparing, for example, a short regional route against a long cross-country one. Reporting both the median and mean lead time additionally guards against a small number of extreme delays distorting the picture of "typical" route performance.

5. Anticipated Findings and Analytical Framing

Because route performance depends on the underlying shipment data, the specific numerical outcomes of this study (e.g., which exact states rank in the bottom 10) are determined once the full dataset is processed through the pipeline described in Section 3. However, the factory network's geography allows several directional hypotheses to be framed in advance and tested against the data:

- **Distance-driven baseline delay:** Customers in regions geographically distant from their assigned factory (for example, Northeastern customers served by Lot's O' Nuts in Arizona, or West Coast customers served by Wicked Choccy's in Georgia) are expected to show higher baseline lead times than customers located near their factory, independent of shipping mode.
- **Single-factory dependency risk:** Because each product division is tied to a small number of factories, states that order heavily from a single division (e.g., a state with concentrated Chocolate demand served only by Lot's O' Nuts or Wicked Choccy's) may show less resilience to delay, since there is no alternate factory to absorb volume during disruptions.
- **Ship mode divergence:** Expedited shipping is expected to substantially compress lead time on long-distance routes but to show diminishing returns on routes that are already geographically short, suggesting that expedited-mode cost premiums may be better targeted at specific long-haul routes rather than applied uniformly.

- **Volume-delay coincidence:** Standard practice in distribution networks is for a small number of high-volume routes to disproportionately drive overall customer experience; the analysis is expected to identify a similarly concentrated set of "priority" routes at Nassau Candy where volume and delay intersect.

These hypotheses form the analytical lens through which the completed dataset should be interpreted, and are intended to be confirmed, refined, or rejected once the full route-level aggregation (Section 3.3) is executed.

6. Dashboard Design: Streamlit Web Application

To operationalize the analysis beyond a static report, findings are delivered through a live, interactive Streamlit dashboard ([app/app.py](https://shipping-route-efficiency-analysis-cww.streamlit.app/)), deployed at <https://shipping-route-efficiency-analysis-cww.streamlit.app/>. The dashboard is organized into a persistent KPI header, a synced filter panel, and five analytical tabs.

6.1 Route Overview

- Route efficiency bar chart and a volume-vs-lead-time scatter plot, allowing high-volume/high-delay routes to be spotted visually.
- Top 10 and Bottom 10 efficient-routes leaderboards.
- A full, sortable route performance leaderboard table.

6.2 Geographic View

- A U.S. choropleth map showing state-level average lead time.
- A regional bottleneck bar chart.
- Automated bottleneck-state detection (states in the top 25% of both shipment volume and average lead time).

6.3 Ship Mode Analysis

- Lead-time comparison across shipping modes.
- Delayed vs. on-time percentage breakdown by ship mode.
- Summary cards calling out the best-performing and slowest shipping modes.

6.4 Route Drill-Down

- A per-route order timeline, distinguishing on-time from delayed shipments.
- A full order-level record table filtered to the selected route.
- An auto-generated analyst recommendation comparing the selected route's performance against the network-wide average.

6.5 Trends

- Monthly average lead time and order volume trend lines.
- Month-over-month movement summaries.
- Automatic identification of peak and lowest volume/lead-time months, paired with analyst recommendations.

Insights and recommendation text throughout all five tabs are generated dynamically from whatever data subset is currently filtered, rather than hardcoded against the full dataset — so the narrative stays accurate as filters change.

6.6 User Capabilities

The dashboard supports interactive filtering so that stakeholders can explore the data without needing analytical expertise. The filter panel appears as a sidebar on desktop and a collapsible expander on mobile:

- Date range filter
 - Region / State selector
 - Ship mode filter
 - Lead-time threshold slider (to dynamically redefine what counts as a "delay")
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7. Project Architecture and Tech Stack

The solution was implemented as a versioned, reproducible project rather than a single script, with the following structure:

```
nassau-candy-shipping-route-efficiency-analysis/
|
|-- app/
|   |-- app.py                # Streamlit dashboard
|   |-- powerbi_ready_dataset_builder.py
|   |-- prepare_dashboard_files.py
|   |-- assets/               # custom CSS theme, images
|
|-- data/
|   |-- Nassau Candy Distributor.csv # raw source data
|   |-- processed/             # cleaned data consumed by the dashboard
|   |-- powerbi_ready/         # exports for Power BI
|
|-- notebooks/
|   |-- 01_project_overview_and_data_loading.ipynb
```

```
| — 02_data_cleaning_and_validation.ipynb
| — 03_feature_engineering.ipynb
| — 04_route_definition_and_aggregation.ipynb
| — 05_efficiency_benchmarking.ipynb
| — 06_geographic_bottleneck_analysis.ipynb
| — 07_ship_mode_performance_analysis.ipynb
| — 08_key_insights_and_recommendations.ipynb
| — 09_export_cleaned_data_for_streamlit.ipynb
|
| — reports/
| — requirements.txt
```

Tech stack: Python; Pandas and NumPy for data processing; Plotly (Express and Graph Objects) for visualization; Streamlit for the dashboard layer, with a custom CSS theme; Streamlit Community Cloud for deployment, connected directly to GitHub so new commits auto-redeploy the live app.

Each notebook in the pipeline corresponds directly to a stage of the methodology described in Section 3 — the split into nine discrete notebooks was a deliberate choice to keep each transformation (cleaning, feature engineering, aggregation, benchmarking, bottleneck detection, ship-mode analysis, insight generation, export) independently reviewable and re-runnable.

8. Future Enhancements

Building on the current dashboard and analysis, the following extensions are identified as natural next steps:

- **Route delay prediction using machine learning**, to move from descriptive to predictive route-risk scoring.
- **Route optimization recommendations**, suggesting alternate factory assignments or shipping modes for chronically underperforming routes.
- **Interactive route maps** that plot factory-to-customer paths directly, rather than state-level aggregates alone.
- **Automated report generation**, so that the insights currently produced interactively in the dashboard can also be exported as scheduled written summaries.
- **Cost vs. lead-time trade-off analysis**, once per-shipment carrier cost data is available, to complement the current descriptive ship-mode comparison with a quantified cost-efficiency frontier.
- **Real-time logistics monitoring**, extending the pipeline from periodic batch analysis to a continuously updating operational feed.

9. Recommendations

Based on the methodology and directional findings outlined above, the following recommendations are proposed for Nassau Candy Distributor:

1. **Prioritize intervention on high-volume, high-delay routes first.** Improvements here yield the largest customer-experience return per unit of operational effort.
2. **Reassess factory-to-region assignment where feasible.** Where a division is manufactured at more than one factory, routing customers to their geographically nearest factory (rather than a fixed default) could reduce baseline lead time.
3. **Target expedited shipping strategically rather than uniformly.** Reserve expedited mode for long-haul routes where it produces the largest lead-time reduction, rather than applying it as a blanket premium option.
4. **Institutionalize continuous monitoring.** Adopt the Streamlit dashboard as a recurring operational tool, reviewed on a regular cadence, rather than a one-time analytical exercise, so that route performance is tracked as the business scales.
5. **Set a formal delay threshold.** Defining an explicit lead-time threshold (via the dashboard's threshold slider) enables consistent tracking of "Delay Frequency" as a standing KPI rather than an ad hoc calculation.

10. Conclusion

This project establishes a structured, data-driven framework for understanding shipping route efficiency at Nassau Candy Distributor. By defining routes at the factory-to-customer-region/state level, engineering shipping lead time as a core metric, and systematically benchmarking routes, regions, and shipping modes against one another, the organization can move from anecdotal, reactive logistics management to proactive, evidence-based decision-making. The accompanying Streamlit dashboard ensures these insights remain accessible and actionable on an ongoing basis, positioning Nassau Candy to reduce delays, control operational cost, and support reliable nationwide delivery as the business continues to scale.

Deliverables Summary

- **Research paper** (this document): exploratory data analysis framework, methodology, and recommendations.

- **Streamlit dashboard:** live, interactive analytics covering route efficiency, geographic bottlenecks, and ship mode comparison.
 - **Executive summary:** a condensed, non-technical brief of findings prepared for government/stakeholder review.
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Project Links

- **Live Dashboard:** <https://shipping-route-efficiency-analysis-cww.streamlit.app/>
 - **GitHub Repository:** <https://github.com/Saswata-pal/shipping-route-efficiency-analysis>
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*Prepared by **Saswata Pal**, Machine Learning Intern, Unified Mentor Pvt. Ltd. , as part of the Unified Mentor Internship Program — Nassau Candy Distributor, Factory-to-Customer Shipping Route Efficiency Analysis.*